

COURSE DESCRIPTION

Course Number	0418-321	Course Title	Operating Systems
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Mansour Abdulaziz
Compulsory	Yes	Prerequisite	0418-223

Textbook

Title	Operating System Concepts
Author	Avi Silberschatz, Greg Gagne, Peter Baer Galvin
Publisher	John Wiley & Sons, Latest Edition

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: Overview of operating systems, operating systems architectures	2
2	Concurrency: Processes and Threads, CPU Scheduling, Process Synchronization, Deadlocks	10
3	Case study (Unix/Linux vs. Windows)	3
4	Memory Management: Swapping, Paging, Virtual Memory	6
5	Storage Management: File Systems Interface, Mass Storage, I/O Systems	10
6	Security: Protection and Security	3
7	Distributed Systems: Networks-based OS, Communication structures and protocols, Network and Distributed File System	3
8	Virtual Machines: Types and Components	2
	Exams	3
	Total	

Lab Schedule

	Topics	Hours
1	Using System Calls and Interfaces	1
2	Computer Booting Sequence	1
3	Shell Implementation	1
4	Kernel Development and Debugging	2
5	Kernel PCB Details	1
6	New System Calls	1
8	Kernel Concurrency and Synchronization tools	1
9	File Systems Implementation	1
10	Distributed File System	1
11	Kernel Process Scheduling	1
12	Disk and Memory Scheduling	1
13	Student Presentation	1
	Total	

Grading

15–20%	Midterm 1
15–20%	Midterm 2
15–20%	Lab
0–15%	Homework Assignments or Quizzes
40%	Final

Course outcomes

Students should be able to:

1. Explain the benefits of a logical layer concept, kernel and user mode concept and how they implemented in modern operating system.
2. Analyze concurrency problems and explain the operating systems support for concurrency.
3. Analyze scheduling algorithms and develop solutions for process scheduling problems.
4. Analyze and understand memory management in operating systems.
5. Understand the difference between serial and parallel devices and the relationship to their virtual devices and how buffering and DMA are used.
6. Understand the need for protection and security and the features and limitations of an operating system used to provide it.
7. Understand the file management systems and how the hardware plays a role in the design.

Relationship between Course Outcomes and Student Outcomes

	SO1	SO2	SO3	SO4	SO5	SO6
CO1	H		L			
CO2	H	M	L			
CO3	H					
CO4	H					
CO5	H					
CO6	H					
CO7	H					